MIL-T-52958B(ME)
31 August 1984

SUPERSEDING
MIL-T-52958A(ME)
2 April 1982

MILITARY SPECIFICATION

TRUCK, FIRE, MINI-PUMPER,

COMMERCIAL CHASSIS

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel engine, four-wheel drive, 250 gallon per minute (gpm) pumping capacity, fire fighting truck having pump and roll capability to provide initial attack operation on a structural and brush/grass fires.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: DRDME-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion Resisting Steel.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubrication Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-P-8116	- Putty, Zinc Chromate, General Purpose.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive Pressure-Sensitive.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty Single Package.
MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
MIL-C-81751	- Coating, Metallic Gramic.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Automotive Vehicles.
FED-STD-595	- Colors.
FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-281	- Automobiles, Trucks, Truck-tractors, and Trailer Dollies, Preservation and Packaging of.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1250	- Corrosion Prevention and Deterioration Control In Electronic Components and Assemblies.
MIL-STD-1474	- Noise Limits for Army Material.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Safety Standards and Regulations.

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations, Title 45, Subtitle A, Part 1201.

Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1	- Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 153 - Zinc Coating (Hot Dip) on Iron and Steel Hardware.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 414 - Aircraft Rescue and Fire Fighting Vehicles.
No. 1901 - Standard for Automotive Fire Apparatus.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The truck shall be four-wheel drive, truck cab and chassis having a gross vehicle weight rating (gvwr) of not more than 15,000 pounds. The truck shall be fitted with a fire truck body, a 250 gpm minimum booster type fire pump, a 250 useable gallon removable water tank, hose beds, a booster hose reel; operator's instrument panel for the fire pump; and such other equipment and accessories as are specified herein.

3.1.1 Standard truck, components, and accessories. Except as specified herein, the truck, components and accessories shall be standard or optional items which meet or exceed the requirements of this specification. Truck,

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components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products, complying with all DOT Federal Motor Vehicle Safety Standards and Regulations applicable, on the date of manufacture.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The truck shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the truck may be newly fabricated from recovered materials to the maximum extent practicable, provided the truck produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the truck.

3.4 Design.

3.4.1 Truck chassis. The truck-chassis shall be a four-wheel drive, front wheel steer, dual rear wheel, conventional type.

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3.4.2 Gross weight rating. The manufacturer's gvw rating for the truck chassis furnished shall be not more than 13,000 pounds. Derating of chassis and components shall not be allowed. The total gross weight of the complete truck shall not exceed the manufacturer's gvw rating when fully equipped as specified herein, including all specified firefighting equipment, serviced with fuel, water and lubricants; loaded with filled water tank; not less than a 800 pound simulated load and a crew of three men.

3.4.2.1 Angle of approach and departure. The angles of approach and departure shall conform to the angle of departure requirements of NFPA No. 414.

3.4.3 Dimensions. The truck overall dimensions including mounted equipment shall be not more than the following:

a. Width - 8 feet (ft)

b. Length - 22 ft

c. Height - 9 ft

3.4.4 Frame. Chassis frame shall be the manufacturer's standard and shall be rated for operation at rated gvw. When operating the fully equipped and loaded truck under conditions specified herein, the frame shall show no evidence of permanent deformation or distortion. Prior to production of the first article truck, the contractor shall obtain certification from the original truck chassis manufacturer that:

a. The chassis frame is suitable for the intended operational use and for the mounting of all required firefighting components.

b. Any frame modification, including heating, or machining, shall be approved and certified by the truck chassis manufacturer.

Certification shall be made available for review by the contracting officer or designated representitive.

3.4.5 Engine. The truck shall be equipped with the manufacturer's current standard liquid cooled, diesel engine rated for fire service use. The engine furnished shall produce the required vehicle performance when operated on fuel conforming to VV-F-800, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of continuous, warranted operation.

3.4.5.1 Governor. Manufacturer's current standard type engine governor shall be furnished, set and sealed to limit engine to maximum recommended operating speed.

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3.4.5.2 Fuel system. Fuel system shall be standard for the chassis furnished. Fuel lines shall be adequately supported and shall be protected against mechanical damage. Construction and mounting shall be such that neither vibration nor tightening of mounting devices will cause leaks to develop.

3.4.5.3 Fuel tank(s). The fuel tank shall be of not less than 19-gallon capacity. Tank and fill piping shall be so placed as to be protected from mechanical injury. Step tank(s) may be furnished, but side running boards are required as specified.

3.4.5.4 Fuel pumps. The fuel system shall be standard for the vehicle furnished. In addition, an auxiliary electric fuel pump with a manually controlled electrical switch shall be furnished and installed in the truck cab. Fuel pumps shall be connected in parallel so that either can be used independently of the other, and each can be tested separately. Each fuel pump system shall be provided with an independent shutoff valve.

3.4.5.5 Fuel filter. The fuel system shall be provided with a fuel filter standard for the vehicle furnished.

3.4.5.6 Air cleaner. An air cleaner shall be furnished.

3.4.5.7 Air pollution control. The truck shall comply with the Federal EPA regulations governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of manufacture.

3.4.5.8 Exhaust system. The exhaust system shall include a muffler and tailpipe. The exhaust shall be located so that entrance of exhaust gages into the truck cab shall be minimized under all conditions of operation. Location of tailpipe shall prevent heating of tires, disturbance of road dust, and any clogging by mud or water. The muffler shall be located not closer than 2 feet from the fuel tank. Where this is not possible shielding will be used to protect the fuel tank from any part of the exhaust system within 1 foot of the tank and insulation and shielding elsewhere, where necessary.

3.4.6 Electrical equipment. The truck shall be provided with a complete 12 volt (V), negative ground, starting and lighting system. Wiring identification shall be consistent throughout the truck. The truck chassis electrical circuits and the truck body electrical circuits shall be separate circuits. The truck body electrical circuits, excluding truck chassis electrical circuits, shall be provided with overload protective devices of the manual or automatic reset, overload-circuit-breaker type. All electrical components shall be water resistant and designed so that their normal functioning will not be impaired by entrance of water during heavy rain, road splash, condensation or from water spray resulting from other causes. The contractor shall certify that corrosion prevention shall be in accordance with MIL-STD-1250, except that in case of conflict between MIL-STD-1250 and MIL-STD-889 with respect to dissimilar metals, MIL-STD-889 shall prevail. All electrical components shall conform to MIL-STD-454, requirement 1.

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3.4.6.1 Ignition system. Ignition system of 12 V potential shall be furnished. Alternator of not less than 140 ampere (amp) rated capacity, and which provides not less than 80 amp direct current (dc) output at normal engine idle speed, shall be furnished.

3.4.6.2 Battery system. A dual 12 V battery system shall be provided. Each battery shall have a cold crank rating of at least 450 amps and a reserve capacity rating of at least 125 min as set forth in SAE J537. The battery units may be mounted on a sliding tray in an enclosed compartment and shall be accessible for testing and servicing. The battery shall be securely held in place with removable locking devices and shall be protected against engine and exhaust heat and road splash. Provisions shall be made for securely locking sliding tray in the retracted position. The battery compartment shall be ventilated and treated against acid fumes and spillage. The battery compartment shall not be mounted on or project onto the side running boards when in the retracted (operating) position. Battery shall be provided with a polarized receptacle for station changing. The receptacle shall be located at the rear of the trucks. The connector shall disconnect when the truck is pulled forward. The receptacle shall be the recessed female type with a spring loaded cover and provided with a matching male plug with protected prongs. Removable battery compartment cover shall be fitted with lifting handles. Finger lift openings are not acceptable. Battery system shall be wired as suggested by NFPA Code 1901 for dual battery system. Battery terminals shall be provided with guards to prevent accidental shorting.

3.4.6.3 Lighting. All truck lights, reflectors and wiring shall be as specified herein. Lights and reflectors shall not be mounted on vertical surfaces of rub rails or bumpers.

3.4.6.3.1 Lights. All lights shall be mounted in such a manner as to prevent being obsecured by riding personnel or mounted equipment. Stoplights and tail-lights shall be flush mounted at the rear of the body. The following additional lighting shall be furnished and shall be provided with overload protection:

a. Two sealed-beam spotlights. The lights shall be installed on the windshield posts with controls extended inside the cab for on-off control and movement through horizontal and vertical angles.

b. Two adjustable, sealed beam, hose pick-up lights, with weatherproof switches, mounted on rear to the outside of the hose bed.

c. Engine compartment lights, arranged to illuminate both sides of the engine, with switch located on cab dashboard.

d. Two automatic back-up lights, not less than 4-inches in diameter, at rear of body.

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e. Pump compartment light or lights arranged to provide illumination of the compartment with door control switch.

f. Pump control panel lights.

g. Two 6-inch sealed beam, red alternating flashing warning lights shall be mounted adjacent to the hose pick-up lights. The rear mounted warning lights shall be operated by means of a single, separate switch in the truck cab. A switch indicating lights shall be provided and shall be marked to indicate function.

h. Two low mounted, red flashing lights shall be mounted on the front of the truck. The warning lights shall be installed on a horizontal plane through the centerline of the headlights. The lights may be installed adjacent to the headlights. The warning lights shall be sealed beam lamps, 6-inches minimum diameter. The low mounted warning lights shall be operated by a seperate switch of the same type switch furnished for the lights in 3.4.6.3.1g. An indicating light shall be located in the truck cab adjacent to the one in 3.4.6.3.1g. to indicate function.

i. Lights in each compartment with individual door control switches.

NOTE: All indicating lights mounted in the cab shall not glare on the truck cab windshield during night operation.

3.4.6.4 Warning light. A cab roof mounted light shall be furnished. Light beams shall be reflected to the front, rear and sides. The sealed beam lamps shall be high intensity, red in color. The light system housing shall be clear. The light shall be provided with a brush guard. The warning light shall be operated by means of a separate switch with red indicator light located in the cab and marked to indicate function of switch.

3.4.6.4.1 Electronic siren, public address, and radio amplifier system. An electronic siren system with selector controls for siren, public address, and radio amplification through an external speaker shall be provided. The system shall be a Federal PA-20A, Dietz 14-450 or Unity SPR73 Model, or equivalent. The speaker shall be mounted behind the grill of the truck. The amplifier and its controls shall be installed inside the vehicle in a location accessible to the driver and assistant driver. A microphone of the magnetic noise cancelling type, connected by a coiled cord, shall be provided and located on the drivers side of the control case. The system shall permit the broadcasting of routine and emergency radio announcements from an on-board vehicle radio to be installed by the Government. The siren shall be capable of being actuated by both a foot operated switch and the system actuator. The amplifier unit shall include volume control and selection of "Radio", "P.A.",

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"Manual", "Yelp", "Wail" and "Hilo" (European) siren modes. A two-way radio audio tap cable permitting incoming two-way radio messages to be rebroadcast through the public address system shall be furnished. The speaker shall be fitted with a weatherproof driver and shall be suitable for installation in the positon specified (see 3.4.6.4).

3.4.7 Heavy-duty cooling system. A heavy-duty cooling system shall be provided. The heavy-duty cooling system shall be of the pressurized type of sufficient capacity to maintain a cooling liquid temperature of not less than 140° F, and to provide proper cooling without overheating or loss of coolant when operated in an ambient temperature range of minus 25° F to plus 125° F as specified.

3.4.7.1 Coolant system. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir.

3.4.7.1.1 Heat exchanger. An auxiliary heat exchanger shall be provided, which shall permit delivery of water from the fire pump to the heat exchanger for cooling of the engine liquid coolant without admixture. The heat exchanger shall not be connected to the cab heater circuit. The discharge water from the heat exchanger shall be piped into the suction side of the fire pump. The size of the heat exchange discharge line shall be large enough to eliminate back pressure on the heat exchanger. The flow of water from the fire pump to the heat exchanger shall be controlled by a valve, with the valve control located on the fire pump operator's control panel.

3.4.7.1.2 Drain cocks. Water drain cocks shall be installed at the low point of the cooling and heat exchanger systems and at any other points necessary to completely drain the systems. The drain cocks shall be of a type that will remain closed under the most severe operation conditions to be encountered by the fire truck. Drain lines shall be furnished and shall discharge at a point not above the bottom of the chassis frame.

3.4.7.2 Drain plug. All drain plugs installed throughout the truck for the purpose of draining lubricants shall be of the permanent magnet type.

3.4.8 Front bumper and towing devices. A standard commercial heavy-duty, full width, front bumper and two hooks shall be provided at the front of the truck. The two front tow hooks shall be mounted to the frame or rigid members which are attached to the frame. Towing ring(s), or comparable U-bolt(s), shall be provided at the rear of the truck.

3.4.9 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. Engine hour meter shall be mounted on the fire pump panel.

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3.4.10 Suspension. Truck shall be equipped with suspension system, with components having a rated capacity at least equal to the load imposed on each member, plus at least 10 percent, measured at the ground, with truck loaded as specified. When suspension is rated at the spring pads, unsprung weight shall be deducted. Truck shall be equipped with hydraulic, heavy duty, double-acting, shock absorbers on the front and rear wheels. The chassis manufacturer's standard auxiliary rear springs shall be furnished.

3.4.11 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.11.1 Transmission. The truck shall be provided with the manufacturer's standard heavy duty 4 speed manual transmission which will meet the requirements specified herein. A drive gear for the fire pump shall be provided. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. When necessary, a caution plate reading: "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKE OFF ENGAGED" shall be installed on the instrument panel when required by the manufacturer.

3.4.11.1.1 Pump drive. A power takeoff (PTO) may be provided to drive the fire pump. The PTO and pump gear ratio shall be selected to provide the required performance. Controls to place the pump in gear shall be located in the chassis cab. Provisions shall be made to protect the pump and PTO from damage by accidently engaging the fire pump or PTO, when the engine is operating.

3.4.11.2 Axles. The axles shall have a rated capacity equal to the load imposed. Rear drive axle shall be provided with positive-traction or limited-slip differential.

3.4.11.3 Steering. Truck shall be equipped with power steering. Mechanical type steering capability shall be available in event of loss of power steering.

3.4.12 Brakes. Brakes shall be standard for the chassis furnished.

3.4.12.1 Service brakes. Truck shall be equipped with power assisted hydraulic actuated, four-wheel, service brakes, using silicon brake fluid conforming to MIL-L-46176.

3.4.12.2 Parking brakes. Truck shall be equipped with a hydraulic actuated, 4 wheel, parking brake.

3.4.13 Wheels, rims, tires and tubes. Truck shall have a single front and dual rear wheels. Rim and tire ratings shall conform to Tire and Rim Association recommendations. Tires shall be tube or tubeless type of not less than 100 level quality with standard commercial mud and snow tread. Steel belted radial tires shall be furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations. All tires furnished on the

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truck shall be of the same size. The rated individual tire capacity at 60 mph highway conditions shall be not less than the maximum wheel loading imposed with truck loaded to rated gvw. Wheel and tire assemblies shall be dynamically balanced at a simulated speeds up to 60 miles per hour (mph). Disk type wheels shall be provided. A spare tire shall be furnished and shipped as an over rail item in the hose bed.

3.4.14 Cab. The cab shall be standard commercial construction and shall provide unobstructed seating for three men. Cab shall be provided with boots for all levers and controls extending to the exterior of the cab. Safety grips shall be provided on each side of the cab to assist personnel in climbing into the truck cab. Fastenings and three pairs of seat belts shall be furnsihed. Dual sun visors shall be furnished. A dome light shall be provided. The seat shall be upholstered over deep air form cushions with first grade Naugahyde or equivalent, suitable for fire service use.

3.4.14.1 Windshield wipers and washers. Truck shall be equipped with the manufacturer's standard windshield wipers and windshield washers.

3.4.14.2 Heater and defroster. The manufacturer's standard hot water heater/defroster shall be provided.

3.4.14.3 Controls and operating mechanism. All controls shall be standard for the vehicle furnished. Instruments and controls shall be identified as to their function. Identification marking shall be the commercial standard for the vehicle furnished.

3.4.14.4 Back-up warning device. The truck shall be provided with a back-up warning device conforming to SAE J944.

3.5 Maintainability.

3.5.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5 and 6.6) shall be adequate to perform the intended function of assisting or conducting maintenance operations.

3.5.2 Ease of maintenance. All major assemblies, including pumps, transmission, hose reel, and similar items, shall be arranged for individual removal and replacement without major disassembly or removal of other components of the truck body. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility.

3.6 Human factors. The characteristics of the truck shall provide for operation by personnel ranging from the small man, clothed, through the large man, artic-coated, in accordance with SAE J833 and J925. Universal symbols in accordance with SAE J1500 shall be used throughout.

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3.6.1 Noise level. The fire truck shall be equipped with all available standard equipment accessories. If the noise level in the area occupied by the operator or personnel normally working with the truck exceeds category D of MIL-STD-1474, the contractor shall perform an evaluation of the feasibility of lowering the noise level to 85dB(A) or less, and provide an analysis and cost estimate of meeting the 85dB(A) limit. If the Government concludes that it is unjustifiable to require the 85dB(A) limitation, the contractor shall provide warning decals and appropriate warnings in manuals in accordance with MIL-STD-1474. The exterior noise level shall not exceed 83dB(A) when measured in accordance with SAE J366.

3.7 Safety. All exposed parts which are subject to high operating temperatures, or which are energized electrically shall be insulated, fully enclosed or guarded. All rotating, reciprocating or moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Fire fighting equipment, when mounted on the truck, shall be located and positioned so as not to present a safety hazard to personnel. Surfaces of the truck which are to be used as walkways, working areas, or steps, shall be provided with non-skid surfaces.

3.8 Performance. The equipped truck, loaded as specified in 3.4.2, shall be capable of performing the following:

a. Parking brakes holding the truck on a 30 percent grade, in both assending and descending positions.

b. Service brakes holding the truck on a 30 percent grade in both accending and descending positions.

c. Service brakes bringing the truck to a stop within a braking distance of 30 feet from a speed of 20 mph on a dry, smooth, level surface free from loose material.

d. Attain a speed of not less than 55 mph on dry level, paved roadway.

e. Accelerating on a level road from a standing start to a speed of 35 mph in 25 seconds.

f. Operation under all conditions of test without component overheating.

g. There shall be no evidence of body distortion, water leakage, tire and body contact, malfunction of components, irregular chassis noise, vibration or sway when subjected to road test of 100 miles over paved roadway at speeds up to 55 mph; 50 miles over graded dirt/gravel roadway at speeds up to 25 mph.

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h. Negotiating side slopes up to 20 percent on a surface reasonably hard and free from loose material.

i. Pump and roll at speeds of 3 to 7 mph while pumping water from the tank to pump and back through the tank fill line at a pump pressure of not less than 100 psi and 20 gpm under the ground condition specified in a. and g.

3.9 Environmental operation requirements.

3.9.1 Temperature. The truck shall be capable of starting and performing in any temperature from plus 125° F to minus 25° F. The contractor shall obtain data from all component suppliers to this effect. The data shall be made available for review by the contracting officer or designated representitive.

3.9.2 Altitude. The truck shall deliver its rated pump capacity when tested to conditions as specified by NFPA 1901. When specified (see 6.2), modification kits shall be provided and installed for utilization of trucks at special elevations.

3.10 Fire pump. The fire pump shall have a manufacturers rating of not less than 250 gpm at 150 pounds per square in (psi) through 20 ft of 3 inch hose with 10 ft suction lift and be a booster type centrifugal pump. The pump drive shall be controllable from the cab. Controls for pumping and pump and roll operations shall be operable from the driver's positon. Positive, manually operated, securing devices shall be provided to positively lock the truck transmission and pump in gear during all pumping operations. The pump impeller shaft shall be of corrosion-resistant steel that has been passivated in accordance with QQ-P-35. The fire pump shall be the manufacturer's standard fire pump. The pump drive, gearing and housing shall be such that at rated capacity and pressure, the temperature rise in the gear oil will not exceed the maximum temperature recommended by the pump manufacturer for continuous operation. When required, oil cooling that will not be damaged by freezing shall be provided. All shafts in the pump gear train shall turn in anti-friction bearings. All pump-shaft glands or mechanical seals shall be adequate for service under operating conditions specified herein. Drain cocks, push-pull or rotary turn type, accessible at the pump operator's position and arranged so that the open and closed position is clearly indicated, shall be provided to drain the pump and all water-carrying lines. Drain cocks shall be of the fire service type and shall be equipped with positive stops. Pump drain lines shall be furnished and shall completely drain and discharge to a point not above the bottom of the chassis frame. All threaded connections shall be NPT and shall conform to FED-STD-H28. The pump installed in the truck shall be tested for two hours at rated conditions. Calculation of pump flow rate shall be made using pump discharge pressure and standard pitot gauge readings.

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3.10.1 Priming pump. The priming pump shall be standard for the truck furnished, operated from the pump control panel.

3.10.2 Suction connections. The fire truck shall be provided with two gated 3-inch American National fire hose coupling thread (NH) female full flow type suction connections; one located on the fire pump operator's control panel, and one at the rear of the vehicle. The valves shall be the pump manufacturer's standard corrosion-resistant, balanced ball type for quarter-turn operation. Each valve shall be operated and locked by means of an individual control handle as a single-handed operation from the operator's panel. The valves shall be fitted with a female swivel and a chained rocker lug plug. The suction connections shall be equipped with a removable and accessible strainer; shall have a machined end to permit hand tight connections; shall provide a seat for suction hose gaskets, and shall be NH. All piping from the suction connection shall be 3 inch inside diameter. Two 3 inch NH female to 2-1/2 inch NH male adaptor shall be provided. Valves shall be inline drop-out type allowing removal by threaded bolts.

3.10.3 Discharge connections. The fire truck shall be provided with one 2-1/2 inch discharge connection located on the pump operator's control panel; two 1-1/2 inch discharge connections located one in each of the 1-1/2 inch cross lay hose beds, and one discharge connection to the booster reel. All discharge connections shall have machined ends to permit hand tight connections. The connection shall provide a seat for discharge hose gaskets, and shall be threaded with NH threads. The 2-1/2 inch connection shall be fitted with a 2-1/2 inch 30 or 45 degree elbow and 2-1/2 x 1-1/2 NH adapter with 1-1/2 inch gasketed cap and chain. The 2-1/2 inch and 1-1/2 inch discharge connections shall be provided with a hose discharge pressure gage located on the fire pump operator's control panel. The discharge valves shall be the manufacturer's standard valve, and shall be balanced, corrosion-resistant, ball type for quarter-turn operation. Valves shall be in line drop-out type allowing removal by threaded bolts. Valves shall be operated and locked by means of an individual control handle as a single-handed operation. The discharge connections shall be provided with drain valves for draining hose lines with the pump discharge gates closed. The push-pull or rotary turn type drain valves shall be of the water service type and shall be equipped with positive stops. Drain-valve controls shall be located adjacent to the discharge connections. Drain lines shall be concealed behind the pump compartment panels and shall discharge at a point not above the bottom of the chassis frame. Each discharge connection, pressure gage, and drain shall be identified by a related numbering system as well as function.

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3.10.4 Relief valve. An automatic relief valve shall be provided for controlling the pressure from all outlets. The location of the relief valve control shall be such that adjustment can readily be made by the operator when standing in normal position, without the use of tools, and in full view of the discharge pressure gage. The relief valve shall be of such design that, when set 10 psi higher than the pump working pressure, there will not be an increase in pump pressure greater than 30 psi when discharge lines are shut slowly. The relief valve shall perform satisfactorily when pumping rated capacity at any pump working pressure over 75 psi.

3.10.5 Fire pump operator's control panel. The truck shall be equipped with a stainless steel fire pump operator's control panel located on the road side of the truck. The panel shall have a nonglare finish. The pump pressure and suction gages shall be not less than 3-1/2 inches diameter, with throttling devices to eliminate pulsating of gages. All gages shall return to zero reading upon equipment shut-down. All extended controls and handles shall be noninterfering, firmly supported, and realinable. Panel shall include at least the following:

a. Pump pressure gage - 30 inch of mercury (Hg) (vacuum) to 600 psi.

b. Pump suction gage - 30 in Hg (vacuum) to 600 psi.

c. Engine tachometer - not less than 4000 revolutions per minute (rpm) range with a provision in the panel to allow checking accuracy with a test revolution counter.

d. Throttle-pull type with latch and microadjustment.

e. Pump pressure by-pass control.

f. All pump discharge and suction valve controls.

g. Tank suction valve control.

h. Primer control.

i. Illumination of all instruments and controls lighting.

j. Engine coolant temperature gage.

k. Engine oil pressure gage.

l. Heat exchanger valve control.

m. Hose discharge pressure gages (0 to 600 psi) for the 2-1/2 and 1-1/2 inch hose connections.

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n. Water tank level gage, electronic type.

o. All other pump panel mounted drains and controls mentioned herein.

p. Discharge valve and throttle handles shall be of black extreme temperature resisting plastic, brass, or stainless steel.

q. Plugged 1/4 inch standard pipe connection(s) with pipe or tubing for connection of vacuum or pressure test gage(s) for parallel testing of pump pressure and suction gages.

r. Instruction plates specified in 3.20.

All controls, valves, drains, and gages shall be identified as to their function and shall have a related identification number. All instruments and switches shall be identified as to their function. Numbers shall be mechanically adhered to the pump panel.

3.11 Body. The truck shall be equipped with a body which shall incorporate a fire pump housing and stowage bin, fire pump operator's control panel, removable water tank, hose bed, equipment cabinets, rear platform, booster hose reel and fire fighting equipment. The body shall be fabricated from aluminum or steel. The body shall be reinforced at all points where hose and ladder clamps and hangers are attached. The body shall be free from rough and sharp edges.

3.11.1 Tool stowage space. Space of sufficient size to accomodate anti-skid chains and emergency reflectors shall be furnished for retaining equipment during truck operation. Fire equipment cabinets shall not be used.

3.11.2 Hose bed. A hose bed shall be provided with the capability of stowing not less than 500 ft of 2-1/2 inch cotton, rubber-lined fire hose. The hose bed interior shall be free of all projections, sharp or rough edges which might chafe or tear fire hose or hinder its placement or removal. The hose bed sides and end shall be flanged and reinforced. The hose bed floor shall be of slatted aluminum or well-seasoned, selected hardwood. The slats shall run the full length of the hose bed, and shall be spaced to provide ventilation for the hose. The hose bed floor assembly shall be removable. Exposed bolt or rivet heads shall be of the oval type. Toe steps shall be furnished; located on the right and left at the rear of body to permit easy access to the hose bed.

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3.11.3 Cross lay hose beds. Two cross lay hose beds shall be located on the top of the pump compartment area. Each bed shall have the capability of stowing not less than 150 ft of 1-1/2 inch double-jacketed, cotton, rubber-lined fire hose. Hose payout shall be provided on each side of the truck with hose rollers installed in the bottom and both sides of each cross lay bed opening to prevent hose interference or snagging. Two 1-1/2 inch swivel type discharge connections shall be provided in the hose bed to allow hose to be removed from either side of the truck. The bottom of the cross lay hose beds shall be of slatted aluminum or well-seasoned hardwood. Discharge gate controls for each hose bed shall be located at pump panel, and shall be suitably identified. The hose beds shall be identified as No. 1 and No. 2. Hose bed No. 1 shall be located toward the front of the vehicle. It is required that the discharge connections be located away from any hose bed to permit hose attachment to, or removal from the connections, without side wall interference.

3.11.4 Equipment compartment. All equipment compartments and compartment doors shall be the manufacturer's standard steel or aluminum box type construction. The cabinets shall be the maximum size practical. Compartments shall be framed of welded or bolted-in structural members for strength and resistance to damage. Compartment doors shall be attached with full length stainless steel piano hinges. All hinged doors shall have spring type door hold-open devices. Each compartment door shall be provided with recessed "D" ring type handles, with two point catch weather resistant latches. Maximum corrosion resistant materials shall be utilized in compartment fabrication and hardware. Doors shall be fully gasketed and weather-stripped with neoprene and all compartment seams shall be completely sealed against water. Drip rails or flanged cover plates shall be provided to deflect rain water from entering compartments. Provisions shall be incorporated to permit flushing and cleaning inside of compartment. Each compartment shall be illuminated, controlled by door switches. Compartment tops and wheelhousing tops shall run continuously on each side of the truck and shall be of antiskid surface, as furnished for running boards and rear platform surfaces. Compartments and compartment doors shall be recessed back from the running board step areas and wheel skirting or shall be protected by a rub rail along each side of the truck. Compartments and compartment doors shall not be flush with the sides of the truck.

3.11.5 Side equipment compartments. The following side compartments shall be furnished:

a. Streetside compartments.

1. One compartment over the rear wheels with door(s).
2. One compartment behind the rear wheels with vertically hinged, forward opening door.

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b. Curbside compartments.

1. One compartment over the rear wheel with door(s).
2. One compartment behind the rear wheels with vertically hinged, forward opening door.
3. One compartment door opposite the streetside pump panel, with vertically hinged, forward opening door, for access to pump, piping, and rear of control panel with ladders removed.

3.11.6 Rear compartment. A rear compartment shall be furnished. The compartment shall be the full width of the hose bed. The hose reel, specified in 3.13, may be mounted in the center of this compartment. Access to the hose reel when mounted in this compartment shall be from the rear of the truck. There shall be sufficient width to permit access to the hose reel for servicing and removal and hose payout. The compartment shall be illuminated and the lighting shall be controlled by a switch.

3.11.7 Rear platform and running boards. The truck shall be provided with a rear platform and running boards. The rear platform shall have a load-carrying capacity of not less than 700 pounds (lbs). The rear platform length shall be equal to the width of the truck. The rear platform shall have rounded corners, and shall be flanged downward at least 2 inches. The side running boards shall extend from the rear of the cab to the side equipment cabinets. Running boards shall have a load-carrying capacity of not less than 350 lbs, and shall be not less than the full width of the body. The rear platform and running boards shall have a steel or aluminum non-slip tread surface. The running boards and rear platform shall be located at the same height from ground level.

3.11.8 Handrails and grabrails. The rear platform shall be provided with two vertical grabrails to assist personnel in climbing onto the rear platform. A horizontal ribbed rubber covered stainless steel handrail shall be provided at the rear end of the hose bed. The handrails and grabrails shall be 1-1/4 inch minimum diameter. All handrails and grabrails shall be of the nontwist type.

3.11.9 Steps. Steps shall be provided to assist personnel in gaining access to various fire fighting body components. As a minimum, one step shall be provided at the pump operator's panel; one step shall be provided on the curbside directly opposite the pump operator's panel, and two steps shall be provided at the rear, one on each side of the rear platform. The steps shall have non-slip surfaces. The maximum height of any step or rear platform shall not exceed 24 inches with the truck loaded as specified.

3.11.10 Buzzer. A buzzer shall be installed in the cab and a weatherproof pushbutton on one of the rear handrail supports to provide for a signaling device from the rear platform to the truck cab.

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3.12 Water tank. A removable water tank of not less than 250 useable gallon capacity shall be provided. The water tank shall not extend beyond the main hose bed side. The water tank shall be fabricated of aluminum or high-strength stainless steel. The stainless steel tank shall be constructed of stainless steel type S30403 per ASTM A 240 and shall be free of mill and weld scale either by a pickling treatment or sandblasting. Longitudinal and transverse surge partitions conforming to NFPA No. 1901 shall be provided in the tank. The water tank shall be of the horizontal type, with flat top, and mounted on the truck frame. The water tank capacity shall be capable of discharging 80 percent of capacity with the fire truck on a 30 percent grade, heading upgrade and also heading downgrade. The top shall be removable, gasketed to prevent leakage, and shall support not less than 200 lbs per square foot of surface area. The top retaining fasteners shall not come in contact with the water. The tank drain shall be located at the low point of the tank to provide complete drainage. The water tank drain shall discharge at a point not above the bottom of the chassis frame. The tank shall have a nonsplash combination overflow and vent pipe directed at the rear of the rear axle, and extending below the bottom of the truck frame.

3.12.1 Tank piping. All piping from the tank to the pump and from the pump to the booster hose reel and cross lay hose bed shall be of corrosion-resistant metal that has been passivated in accordance with QQ-P-35 or galvanized with ASTM A 153 to an F90 coating designation. The pipeline from the pump may incorporate flexible, reinforced, noncollapsible connections to take up vibration. A quick-opening tank to pump shutoff valve shall be provided with control extended to the operating side of the truck, and a drain valve located at the lowest point with control extended to the operating side of the truck. The tank piping shall provide for discharging the water in the tank through the fire pump discharge manifold at a flow rate of not less than 250 gpm. The tank suction pipeline, valves, and fittings shall withstand not less than 120 psi working pressure. The tank to pump line shall be provided with a check valve. The piping between the pump, booster reel and cross lay hose beds shall be not less than 1-1/2 inch and shall incorporate 1-1/2 inch quick opening valves with controls extended to the pump operator's panel. A complete piping diagram shall be provided. All drain lines shall discharge at a point not above the bottom of the chassis frame.

3.12.2 Tank mounting. The tank shall be so mounted to the chassis that it can be removed without removing the truck body.

3.12.3 Tank filler opening. The tank shall have an accessibly located filler opening with a tight, gasketed, hinged cap having a spring-type hold-down catch. The tank fill opening shall be provided with a removable strainer with openings that are not more than 1/4-inch in area and not less than 13/64-inch in area. The tank fill opening shall not be located in the hose bed. A line from the pump to the tank shall be provided for filling the tank, with a valve control located on the pump panel.

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3.12.4 Tank level gage. Two water tank level gages shall be furnished and located on the fire pump operator's control panel and in the truck cab. The gages shall be of the electronic type.

3.13 Booster hose reel. A booster hose reel shall be provided and installed on the truck. The reel may be mounted in the compartment specified in 3.11.6 or over the pump compartment. The reel shall be an electrical rewind type. The rewind control switch shall be a push type, spring loaded, button. One electric rewind control switch for rear hose reel and one switch on each side of the truck for top mounted reels shall be installed and located near the hose reel. The switches shall be suitably identified. The hose reel shall be equipped with 150 ft, in three 500 ft lengths, of 1 inch, rubber chemical hose, jointed with smooth full-flow type couplings. A 1 inch hose nozzle of the combination straight stream and adjustable fog type shall be furnished, Akron style 1710 or equal. A bracket shall be provided for holding the nozzle. The hose reel shall have a friction brake payout to prevent over-travel. The hose reel shall also have provisions for hand-crank rewind. The drain line shall discharge to a point not above the bottom of the chassis frame. Hose reels shall provide for payout of the hose from either side of the truck. Hose rollers shall be provided on both sides and the bottom to prevent hose interference.

3.14 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Outside rearview mirrors. The truck shall have exterior mounted rearview mirrors on both the right and left sides of the cab. Combination (flat and convex) mirrors enclosed in a common holder shall be furnished. Stick-on type convex mirrors are not acceptable. Mirrors shall be of the fold back west coast type.

b. Tachometer.

c. Ammeter.

d. Fuel gage.

e. Oil pressure gage.

f. Engine temperature gage.

g. Speedometer, with recording odometer.

h. Dual sun visors.

i. Door mounted armrests.

j. Manufacturer's standard horn.

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k. Pump discharge pressure gauge.

l. Water tank liquid level indicator.

3.15 Communication equipment. When specified (see 6.2), a radio communication system, or a space approximately 1-1/2 ft long x 1-1/2 ft wide x 1 ft high for installation of communication equipment, shall be provided.

3.16 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the truck shall be certified to SAE J551.

3.16.1 Radio noise suppressors. The truck chassis shall be equipped with radio noise suppressors conforming to SAE J552.

3.17 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations. Pressure lubrication shall not damage seals or other parts. Unless otherwise specified (see 6.2), all parts requiring lubrication shall be lubricated prior to delivery with lubricants conforming to MIL-L-2104 or MIL-L-2105, as applicable, and a tag to show the type and temperature rating of the lubricant used shall be placed in the cab on the transmission control arm.

3.18 Painting. Unless otherwise specified (see 6.2), the truck cab, chassis body and other equipment, as applicable, shall be cleaned, treated, primed, and painted three coats gloss lime-yellow, color No. 13670 of FED-STD-595 in accordance with the manufacturer's standard commercial practice, but not less than equal to MIL-T-704. The tread plates shall be painted aluminum color. The vehicle shall be rust-proofed in accordance with FED-STD-297.

3.19 Identification plate. An identification plate shall be furnished. The contractor shall securely affix it to each fire truck on the forward inside wall of the cab, with nonferrous screws or bolts. The following nomenclature shall be stamped in the space provided on the identification plate: "TRUCK, FIRE FIGHTING, 250 GPM", the vehicle registration number provided by the contracting office, and the assigned National Stock Number (see 6.7).

3.20 Instruction plates. The following instruction plates shall be provided:

a. A plate indicating the fire pump's tested delivery rate in gpm at 150, 200, and 250 pounds per square inch gage (psig) pressure, with manufacturer's specified engine speeds shall be affixed to the fire pump operator's control panel.

b. All pumping controls, valves, instruments, and such fire truck operating controls whose function is not obvious shall be provided with suitable instruction plates designating function and operation.

Plates shall conform to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material.

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3.21 Contractor furnished equipment. The following equipment shall be provided:

a. Two 8 ft lengths of 3 inch hard suction hose with 3 inch NH long-handled couplers on female ends and 3 inch NH rocker lug couplers on the male ends. The suction hose shall conform to the requirements of NFPA No. 1901. One 3 inch strainer shall be provided, mounted and stored on the step plate at the pump panel.

b. Two hose troughs shall be provided. The troughs shall be fabricated of the same material as that used for the body. The cross sectional design shall permit ventilation between the hose and the trough. All metal quick-release fastenings shall be provided adjacent to both ends of the hose to release the two hoses through a single operation. The fasteners shall limit movement of the hose during travel of the truck. The hose shall be mounted on the street side of the truck over the cabinets.

c. One 12 ft straight aluminum ladder. The ladders shall be mounted with quick release fastening on the curbside of the truck over the cabinets.

3.22 Servicing and adjusting. Prior to acceptance of the truck by the Government inspector, contractor shall service and adjust each truck and its mounted equipment for operational use, including at least the following: Focusing of lights; adjustment of engine; adjustment of electrical and brake system; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear and mounted equipment; servicing of cooling system with a solution of anti-freeze and water in equal parts, and servicing of windshield washer reservoir with water and appropriate additives.

3.23 Workmanship.

3.23.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Before assembling steel fasteners (screws, bolts, washers, etc.) in aluminum, a zinc chromate inhibitor or a metallic ceramic coating should be used, such as a zinc chromate paste in accordance with MIL-P-8116 or a metallic ceramic coating in accordance with MIL-C-81751. All parts shall be free from rust, toolmarks, pits and other injurious defects. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are necessary or where they are not detrimental to safety.

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3.23.2 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

3.23.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided and all bolts, nuts, and screws shall be tight. The use of sheet metal screws shall not be permitted.

3.23.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of proper shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.5 Welders and welding.

3.23.5.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

Certification shall be made available for review by the contracting officer or designated representitive. Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes.

3.23.5.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16-inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the truck shall be at the option of the contractor.

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3.23.5.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or splatter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article truck shall be examined as specified in 4.5.1. The examination shall be conducted after all of the tests specified in 4.3.2 have been completed. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article truck shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.3.3 Inspection failure. Failure of a first article truck to meet any requirement specified herein during and as a result of the examination and tests specified shall be cause for rejection of the first article trucks and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each truck shall be examined for defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each truck and component shall be tested as specified in 4.5.2.1, 4.5.2.1.1, 4.5.2.1.2 and 4.5.2.2. The road test specified in 4.5.2.2 shall be reduced by one half the required milage. Failure of any test shall be cause for rejection.

4.5 Inspection procedure. Prior to examination or tests, the truck shall be lubricated with military oil and greases for use at the ambient temperature at the inspection site.

4.5.1 Examination. The truck shall be examined as specified herein for the characteristics listed in table I.

TABLE I. Examination schedule (requirements).

Number	Characteristic	Requirement Paragraph
101.	Truck, components, and accessories not as specified.	3.1
102.	Material not as specified	3.3
103.	Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
105.	Contractor does not have documentation available for identification of material, material finishes or treatments.	3.3.1.2
106.	Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the truck.	3.3.2
107.	Chassis components not as specified.	3.4.1
108.	Weight rating not as specified.	3.4.2
109.	Angles of approach and departure not as specified.	3.4.2.1
110.	Truck overall dimensions not as specified.	3.4.3
111.	Frame not as specified.	3.4.4
112.	Engine not as specified.	3.4.5
113.	Governor not as specified.	3.4.5.1
114.	Fuel system not as specified.	3.4.5.2
115.	Fuel tank(s) not as specified.	3.4.5.3
116.	Fuel pumps not as specified.	3.4.5.4
117.	Fuel filter not as specified.	3.4.5.5
118.	Air cleaner not as specified.	3.4.5.6

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TABLE I. Examination schedule (requirements). - Continued

Number	Characteristic	Requirement Paragraph
119.	Air pollution control not as specified.	3.4.5.7
120.	Exhaust system not as specified.	3.4.5.8
121.	Electrical equipment not as specified.	3.4.6
122.	Ignition system not as specified.	3.4.6.1
123.	Battery system not as specified.	3.4.6.2
124.	Lighting not as specified.	3.4.6.3
125.	Lights not as specified.	3.4.6.3.1
126.	Warning light not as specified.	3.4.6.4
127.	Electronic siren, public address, and radio amplifier system not as specified.	3.4.6.4.1
128.	Heavy duty cooling system not provided as specified.	3.4.7
129.	Coolant system and indicators not as specified.	3.4.7.1
130.	An auxiliary heat exchanger not provided as specified.	3.4.7.1.1
131.	Drain cocks not installed as specified.	3.4.7.1.2
132.	Drain plug not as specified.	3.4.7.2
133.	Front bumper and towing devices not provided as specified.	3.4.8
134.	Engine hour meter not furnished as specified.	3.4.9
135.	Suspension not as specified.	3.4.10
136.	Vehicle not furnished with an automatic transmission as specified.	3.4.11.1
137.	Power takeoff not as required.	3.4.11.1.1
138.	Axles not as specified.	3.4.11.2
139.	Steering not as specified.	3.4.11.3
140.	Brakes not as specified.	3.4.12
141.	Service brakes not as specified.	3.4.12.1
142.	Parking brakes not as specified.	3.4.12.2
143.	Wheels, rims, tires, and tubes not as specified.	3.4.13
144.	Cab not as specified.	3.4.14
145.	Windshield wipers and washers not as specified.	3.4.14.1
146.	Heater and defroster not as specified.	3.4.14.2
147.	Controls and operating mechanism not as specified.	3.4.14.3
148.	Back-up warning device not as specified.	3.4.14.4
149.	Maintenance support not as specified.	3.5.1
150.	Safety not as specified.	3.7
151.	Fire pump not as specified.	3.10
152.	Suction connections not as specified.	3.10.2
153.	Discharge connections not as specified.	3.10.3
154.	Relief valve not as specified.	3.10.4
155.	Fire pump operator's control panel not as specified.	3.10.5
156.	Body of truck not as specified.	3.11

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TABLE I. Examination schedule (requirements). - Continued

Number	Characteristic	Requirement Paragraph
157.	Tool stowage space not as specified.	3.11.1
158.	Hose bed not as specified.	3.11.2
159.	Cross lay hose beds not as specified.	3.11.3
160.	Equipment compartment not as specified.	3.11.4
161.	Side equipment compartments not as specified.	3.11.5
162.	Rear compartment not as specified.	3.11.6
163.	Rear platform and running boards not as specified.	3.11.7
164.	Handrails and grab rails not as specified.	3.11.8
165.	Steps not as specified.	3.11.9
166.	Buzzer not as specified.	3.11.10
167.	Water tank not as specified.	3.12
168.	Tank piping not as specified.	3.12.1
169.	Tank mounting not as specified.	3.12.2
170.	Tank filler opening not as specified.	3.12.3
171.	Tank level gage not as specified.	3.12.4
172.	Booster hose reel not as specified.	3.13
173.	Accessories and equipment not furnished as specified.	3.14
174.	Truck chassis not equipped with radio noise suppressors as specified.	3.16.1
175.	Lubrication not as specified.	3.17
176.	Identification plate not as specified.	3.19
177.	Instruction plates not as specified.	3.20
178.	Fire truck equipment not furnished as specified.	3.21
179.	Servicing and adjusting not as specified.	3.22
180.	Metal fabrication not as specified.	3.23.1
181.	Castings not as specified.	3.23.2
182.	Bolted connections not as specified.	3.23.3
183.	Riveted connections not as specified.	3.23.4
184.	Welding not as specified.	3.23.5.2
Minor		
201.	Treatment and painting not as specified.	3.18

4.5.2 Tests.

4.5.2.1 Pumps. The pumps shall be tested for two hours as specified by NFPA 1901. Data shall be collected every 15 minutes of the test. All gages shall be checked for accuracy of 10 percent of total gage range.

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4.5.2.1.1 Hydrostatic. The entire pump, including the suction chamber and all piping, shall be tested to a hydrostatic pressure of 150 psi for 10 minutes. Nonconformance to 3.10 shall constitute failure of this test. This test shall be conducted after all tests specified in 4.5.2.2 have been completed.

4.5.2.1.2 Test for ambient temperature and altitude conditions. When specified (see 6.2), the contractor shall perform a pumping test of the first article under actual or simulated conditions of the specified altitude above sea level at an ambient temperature of 125° F. When pumping at heights above 2000 feet altitude is required, the contractor shall include certification in the first article report that the truck will perform as specified 3.4.7.1.1. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.2 Road tests. The loaded truck, including filled water tank, filled gasoline tank, lubricants, a 1500 pound evenly distributed simulated load, and a crew of three men, weighing not less than 175 pounds each or equivalent dead weight, shall be tested as follows:

a. 100 miles over paved roads at maximum traffic speeds up to 55 mph.

b. 50 miles over graded dirt, gravel and unpaved roads at speeds up to 25 mph.

c. Service brakes shall be applied during runs a. and b. above.

d. All lights, siren, horn, speedometer, hourmeter, controls, and instruments pertaining to the truck engine shall be tested.

e. The airbrake system shall be tested for conformance to 3.4.12.2.

Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.3 Electromagnetic interference. The truck shall be tested as specified in SAE J 551. In lieu of tests, the contractor may furnish a certification that the truck chassis meets the requirements, together with a list of the suppression devices installed. Nonconformance to 3.16 shall constitute failure of this test.

4.5.2.4 Noise level. The truck shall be tested in accordance with MIL-STD-1474. Typical duty cycle tests shall be used. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing and marking of the truck and equipment shall be examined to determine compliance with the Quality Assurance Provisions of MIL-STD-281 and MIL-E-10062, as applicable.

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5. PACKAGING

5.1 Preservation. The degree of preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete truck, with equipment shall be preserved in accordance with MIL-STD-281, level A, (mobile), with the following additions and exceptions.

5.1.1.1 Engine. Engine and engine accessories shall be preserved in accordance with MIL-E-10062, type I, method I, level A.

5.1.1.2 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-STD-281, except the antifreeze shall conform to MIL-A-46153.

5.1.1.3 Equipment. Equipment shall be secured as provided for, except exposed equipment subject to pilferage, damage or loss shall be prepared as "On vehicle equipment" in accordance with MIL-STD-281.

5.1.2 level B. Each complete truck shall be preserved in accordance with MIL-STD-281, level B (mobile), and with the additions and exceptions as specified for level A.

5.1.3 Level C. Each complete truck and all equipment issued with the truck shall be suitably preserved to afford protection against deterioration and damage from the contractor to the initial destination. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or dessicated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrappings, pack compartmentization, or cartonizing of the individual item. Exposed equipment subject to easy pilferage or loss shall be placed in secured compartments, truck cab or in wooden boxes. Wooden boxes, when used, shall be secured to the truck with metal strapping and in a manner to prevent damage to the painted surfaces of the truck. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153 and a tag attached indicating the temperature to which the cooling system can be subjected before damage. The crankcase and all other lubrication reservoirs shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used and its grade and temperature range. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and other servicing required prior to placing the truck in operation.

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5.2 Packing (see 6.8). Each complete truck, preserved as specified in 5.1 shall not require packing.

5.3 Marking (see 6.8). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The truck is intended for initial attack in combating structural fires.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. When a first article is required for inspection and approval, and numbers of trucks required (see 3.2).

c. When altitude modification kits are required (see 3.9.2).

d. Whether communication equipment or space is required (see 3.15).

e. When electromagnetic interference requirements shall be other than as specified (see 3.16).

f. When lubrication other than as specified is required (see 3.17).

g. When truck exterior is to be treated and painted other than as specified (see 3.18).

h. When the welding process will not be at the option of the contractor (see 3.23.5.2).

i. When altitude and high temperature pumping test shall be performed on the first article model (see 4.5.2.1.2).

j. Level of preservation required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more truck assemblies. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

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6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Ambient temperature. The temperature of the atmosphere measured at the intake side of the engine cooling radiator.

6.4.2 Excessing surging. Excessing surging shall be defined as any movement of the water or fuel that would cause the operator of the truck to lose control during acceleration, deceleration or any maneuver of the vehicle.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each truck.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of trucks.

6.7 Identification information. The contracting officer should arrange to furnish vehicle registration numbers and the assigned National Stock Number.

6.8 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the truck covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC group 91.

6.9 Marking. Levels of packing have not been included as they are not appropriate for this item. For purpose of preservation/packing level marking, the marking shall be as required by the customer.

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Army - ME

Review activity:
Army - CE

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-52958B(ME)	Truck, Fire, Mini-Pumper, Commercial Chassis

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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